

Question 2: Testing

Assignment

Marktplaats has decided on implementing a product feature that we think will drive the number of Car ads with leads: How would you ideally set up the test whether this initiative has improved this metric?

- Why do you choose this method?
- What are the risks / challenges?
- Use the dataset attached to examine the test groups:
 - What is the result of your analysis?
 - What insights did you gain?

1. Business Question and Test Design

Primary KPI

Share of Car Ads With at Least One Lead.

```
has_any_lead = telclicks > 0 OR bids > 0 OR n_asq > 0 OR  
webclicks > 0
```

Test Design

Use a **randomized A/B test** to answer whether the new product feature increases the share of car ads that receive at least one lead.

Why this method:

- Random assignment is the cleanest way to estimate whether the feature caused a change in lead rate.
- It is better than a before/after comparison, which can be affected by seasonality, buyer demand, or changes in ad supply.

Preferred randomization level: ad-level, assuming the feature affects individual ads and does not strongly change seller behavior across ads. If the feature is seller-side, seller-level randomization may be safer.

Treatment Assumption

- I assume B is the treatment group and A is control.
- If the assignment were reversed, the product interpretation would reverse too.

1.1 Design Risks

Before treating the A/B gap as causal, I would validate:

1. **Assignment integrity:** confirm A/B assignment was random, consistently applied, and close to the planned allocation. Any assignment bug or sample-ratio mismatch could bias the result.
2. **Group comparability:** check whether A and B are balanced on key ad and car attributes such as price, mileage, age, seller type, and placement timing. The dataset shows B differs from A on some dimensions, so part of the lift could reflect mix rather than product impact.
3. **Platform side effects:** monitor whether the feature shifts leads away from control ads or increases low-quality leads, duplicate leads or complaints.

```
In [1]: import importlib
import sys
from pathlib import Path

MODULE_DIR = Path("modules").resolve()
if not MODULE_DIR.exists():
    MODULE_DIR = Path("assignments/cars/modules").resolve()
if str(MODULE_DIR) not in sys.path:
    sys.path.insert(0, str(MODULE_DIR))

from IPython.display import display

import functions as helpers

helpers = importlib.reload(helpers)
pd = helpers.pd

pd.set_option("display.float_format", "{:,.3f}".format)
pd.set_option("display.max_columns", 80)
helpers.set_plot_style()
```

2. Data Quality Checks

Before comparing outcomes, I check whether the lead metrics and ad identifiers are reliable enough for analysis:

- Lead metrics are clean enough for analysis: almost no missing values and no negative values.
- Missing lead metrics are minimal: 9 rows have missing values in at least one lead column, so treating missing lead counts as zero has negligible impact.
- Duplicate ad IDs are negligible: 10 rows, checked later as a sensitivity test.

```
In [2]: raw_data = helpers.read_ab_test_data(helpers.DATA_PATH)
ab_data, excluded_data = helpers.prepare_ab_data(raw_data)
segmented_data = helpers.add_segments(ab_data)

print(f"Rows: {len(raw_data):,}")
```

Rows: 183,062

2.1 Lead Metric Quality

Checks whether lead-event columns have missing, negative, or extreme values.

```
In [3]: display(helpers.style_table(helpers.lead_metric_quality(raw_data)))
```

	Metric	Missing Rows	Missing %	Negative Rows	Zero Rows	Max Value
0	Tel Clicks	9	0.000	0	125,624	162
1	Bids	9	0.000	0	152,145	124
2	Questions	0	0.000	0	122,577	282
3	Web Clicks	9	0.000	0	85,122	410

2.2 Missing Values

Shows which optional columns have missing values.

```
In [4]: display(helpers.style_table(helpers.column_quality_summary(raw_data).query('src_ad_id < 10')))
```

	Column	Dtype	Missing Rows	Missing %	Unique
0	Energy Label	str	49,384	0.270	7
1	Body Type	str	17,851	0.098	8
2	Seats	Int64	11,743	0.064	16
4	Doors	Int64	5,375	0.029	9
5	Model	str	925	0.005	2,447
6	Emissions	Int64	191	0.001	356
7	Bids	Int64	9	0.000	42
8	Tel Clicks	Int64	9	0.000	65
9	Web Clicks	Int64	9	0.000	154
10	Price	Int64	7	0.000	7,522

2.3 Duplicated Ad IDs

There are 5 duplicated `src_ad_id` values, covering 10 rows. The summary below is enough to show the issue: repeated IDs refer to rows with different car attributes, so they are not full duplicate rows.

```
In [5]: display(helpers.style_table(helpers.duplicated_ad_id_differences(raw_data)))
```

	Ad ID	Rows	Groups Seen	Varying Columns
0	1011548826	2	A, B	carrosserie, photo_cnt, aantaldeuren, bouwjaar, emissie, energielabel, brand, vermogen, webclicks, model, price, group
1	1027581466	2	A, B	carrosserie, photo_cnt, aantaldeuren, bouwjaar, emissie, energielabel, brand, vermogen, webclicks, model, price, group
2	1045227572	2	A, B	carrosserie, bouwjaar, emissie, energielabel, brand, vermogen, webclicks, model, price, group
3	1054003502	2	A	telclicks, carrosserie, photo_cnt, aantaldeuren, bouwjaar, emissie, energielabel, brand, vermogen, model, price
4	1072325510	2	A, B	carrosserie, photo_cnt, aantaldeuren, bouwjaar, emissie, energielabel, brand, vermogen, webclicks, model, price, group

The paired duplicate-ID rows differ on car attributes. Decimal-looking ad IDs are preserved as text, and these duplicates are exact repeated IDs rather than a truncation issue. I keep them in the primary analysis because they are only 10 rows, then test sensitivity later by excluding them.

2.4 Undocumented Fields

- The dataset contains some fields not documented in the assignment schema.
- Some undocumented fields are useful for balance and segment checks.
- `l2` is excluded from these checks because its meaning is unclear.

3. Group Assignment and Balance Checks

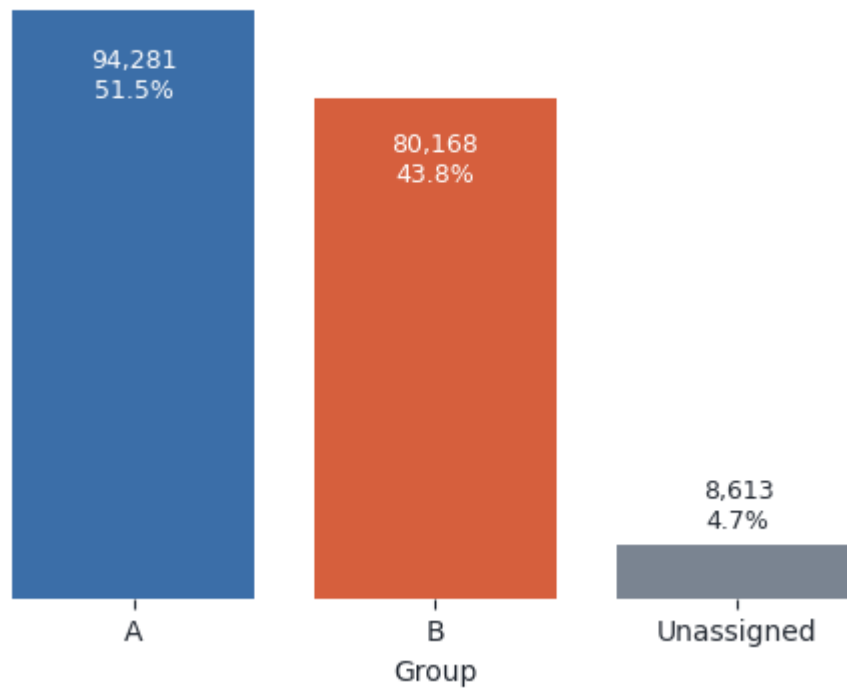
These checks show whether the A/B groups are suitable for causal interpretation before reading the result.

3.1 Group Assignment Split

- Raw groups: A, B, and Unassigned.
- 95.3% of rows are assigned to A or B; 4.7% are unassigned and excluded from A-vs-B outcome analysis.
- If the intended design was 50/50, the A/B split should be checked for sample-ratio mismatch before causal interpretation.

```
In [6]: helpers.plot_group_sizes(raw_data)
helpers.plt.show()
```

Rows by experiment assignment



The dataset has 174,449 valid A/B rows and 8,613 unassigned rows. If the intended split was 50/50, the A/B allocation should be checked with the experiment owner before interpreting the result causally.

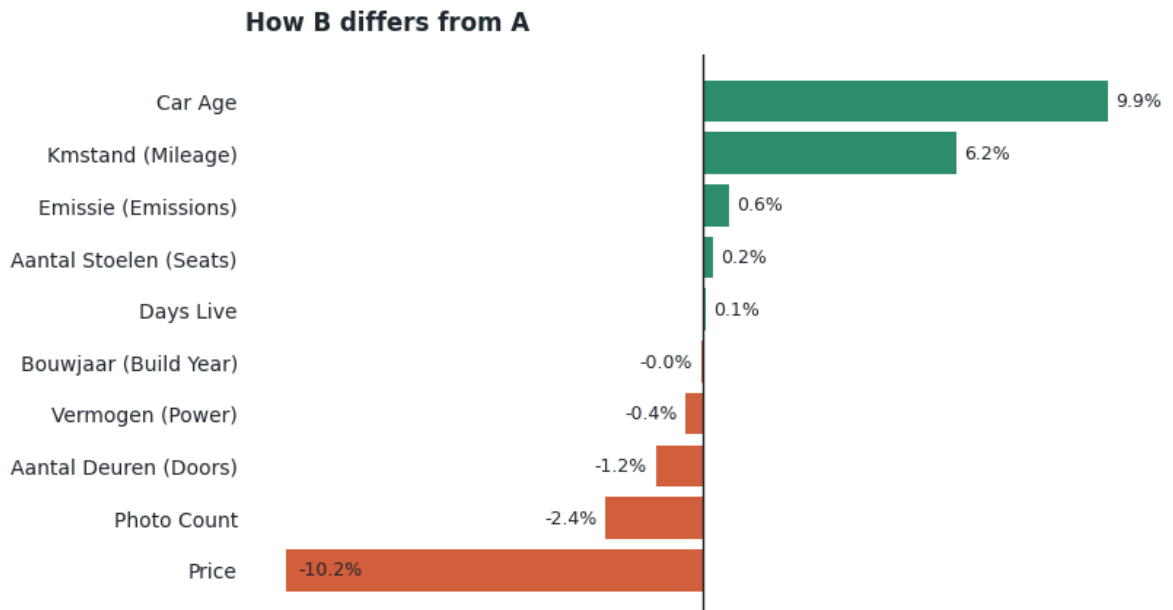
3.2 Group Balance

A and B groups differs on key numeric car/ad dimensions, especially price, car age, and mileage.

```
In [7]: display(helpers.style_table(helpers.numeric_balance_summary(ab_data)))
```

group	Dimension	A	B	Diff B-A	% Diff	Std Diff
0	Price	106,457.104	95,563.256	-10,893.849	-0.102	-0.012
1	Mileage	140,764.544	149,516.859	8,752.315	0.062	0.094
2	Days Live	27.624	27.646	0.022	0.001	0.001
3	Photos	15.742	15.366	-0.376	-0.024	-0.058
4	Build Year	2007.560	2006.721	-0.839	-0.000	-0.155
5	Doors	4.270	4.220	-0.049	-0.012	-0.050
6	Seats	4.735	4.747	0.012	0.002	0.013
7	Emissions	129.123	129.950	0.827	0.006	0.012
8	Power	85.787	85.420	-0.367	-0.004	-0.009
9	Car Age	8.440	9.279	0.839	0.099	0.155

```
In [8]: helpers.plot_numeric_balance(ab_data)
helpers.plt.show()
```



4. Primary Results

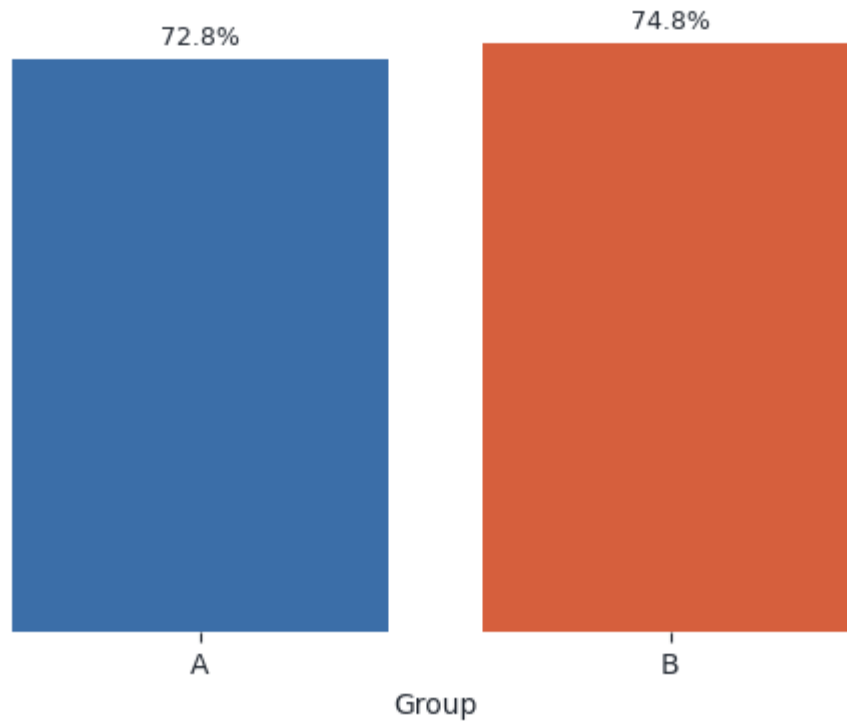
4.1 Result Summary

- Assuming B is the group that received the feature, B has a 74.8% lead rate vs 72.8% for A.
- Difference: +2.0 percentage points; statistically significant ($p < 0.001$).
- B also has more total leads per ad.
- Caveat: groups are not perfectly balanced, so some of the gap could reflect group mix rather than only the product feature.
- Recommendation: B is directionally positive, but investigate both the assignment imbalance and segment mix imbalance before rollout; also confirm which group received the feature.

```
In [9]: outcomes = helpers.lead_outcome_summary(ab_data)

helpers.plot_lead_rate(outcomes)
helpers.plt.show()
```

Share of ads with at least one lead



4.2 Statistical Inference

- The chart shows the primary KPI directly.
- The table below focuses on lift, confidence interval, and statistical inference.
- I use a **two-sided** two-proportion z-test. Even though the feature is expected to help, a product change could theoretically also hurt leads, so I test for a difference in either direction.
- Given the group assignment and balance concerns above, I interpret statistical significance as evidence that the raw A/B gap is unlikely to be sampling noise, not as causal proof by itself.
- Absolute lift: B minus A, in percentage points.
- Relative lift: absolute lift divided by A lead rate.

```
In [10]: display(helpers.lift_summary_table(ab_data).style.hide(axis="index"))
```

Metric	Value
Absolute lead-rate lift	2.0 pp
95% CI for absolute lift	1.6 to 2.4 pp
Relative lift vs A	2.8%
p-value	<0.001
Significance level	5%

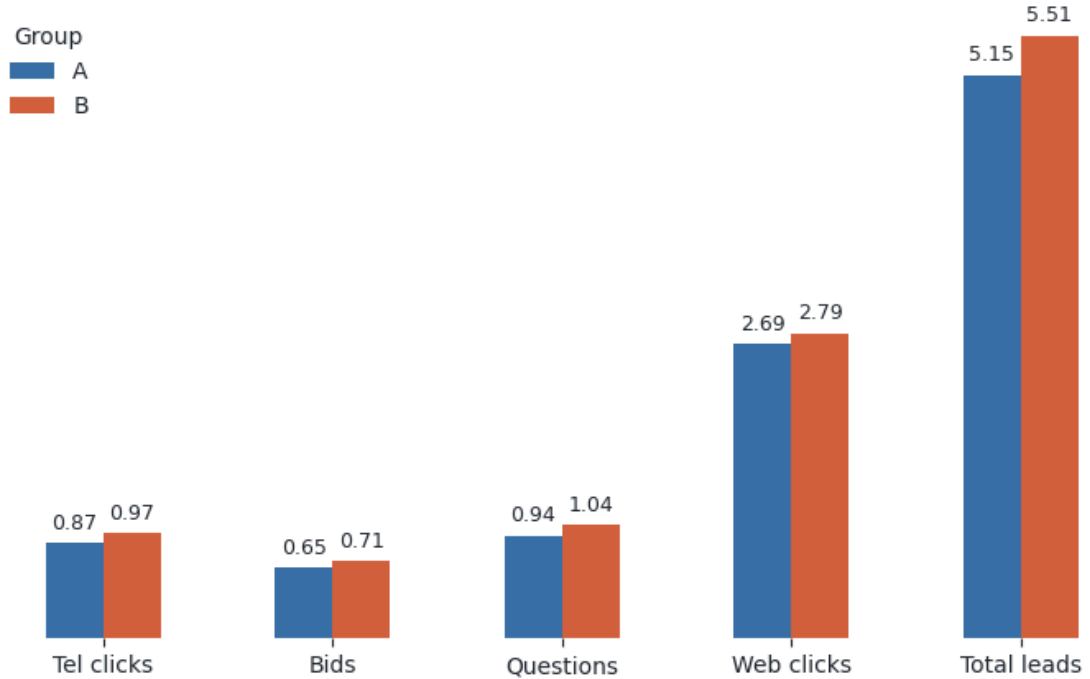
5. Post-Inference Checks

5.1 Lead Volume Pattern

- B also has higher average lead volume per ad.
- Channel checks show whether the lift is broad or concentrated in one lead type.

```
In [11]: helpers.plot_average_leads_by_channel(outcomes)
helpers.plt.show()
```

Average leads per ad by channel



- B is higher on every lead channel and total leads per ad.
- The lift is spread across channels, not driven by one large channel spike.

5.2 Duplicate-ID Robustness

Purpose: verify that the duplicate-ID issue found during data readiness does not drive the primary conclusion.

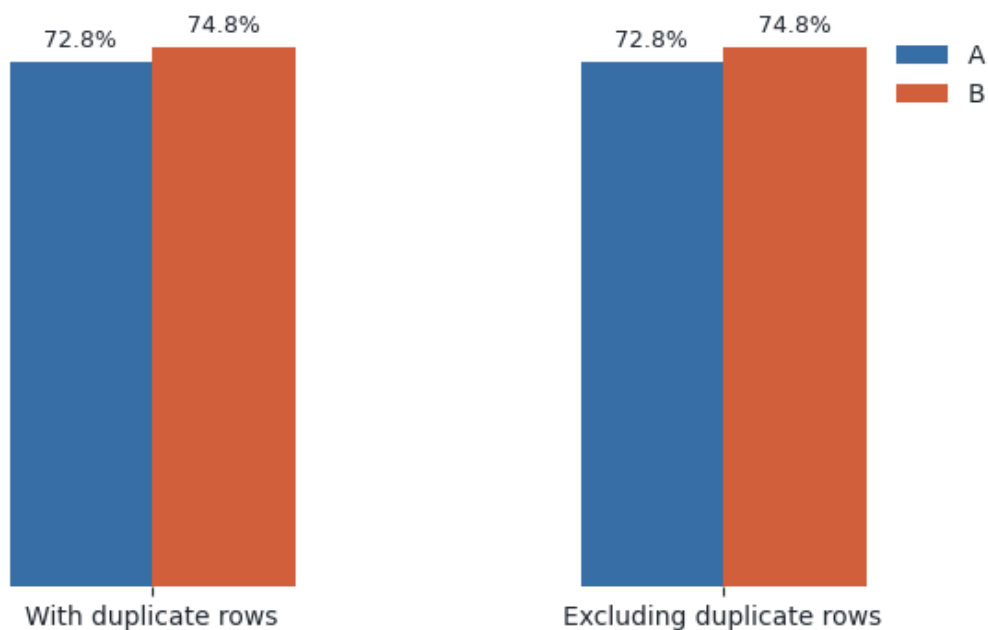
```
In [12]: display(helpers.style_table(helpers.duplicated_ad_id_sensitivity(ab_data)
```

	Metric	With Dup IDs	Excl Dup IDs
0	Rows	174,449	174,439
1	Rows Dropped	0	10
2	% Rows Dropped	0	0.000
3	A Lead Rate	0.728	0.728
4	B Lead Rate	0.748	0.748
5	Abs Lead-Rate Lift	0.020	0.020
6	Relative Lift vs A	0.028	0.028

Excluding duplicated `src_ad_id` rows does not materially change the A/B lead-rate comparison. The result is robust to this specific data-quality issue.

```
In [13]: helpers.plot_duplicated_ad_id_sensitivity(ab_data)
helpers.plt.show()
```

Duplicate-ID sensitivity



Removing duplicated ad IDs leaves the A and B lead rates essentially unchanged. The duplicate-ID issue is worth documenting, but it does not drive the main result.